



JESRYL JADE Y. LAZAGA

Embedded Software Engineer

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SUMMARY

Passionate Embedded Software Engineer with hands-on expertise in designing and developing firmware for microcontrollers and embedded Linux systems. Proficient in C, C++, Python, and Bash, with extensive experience across a variety of microcontrollers. Skilled in implementing communication protocols, including UART, SPI, I2C, CAN, BLE, and LoRaWAN, and experienced in real-time systems using FreeRTOS. Adept at bridging hardware and software, from interfacing sensors and actuators to building scalable IoT solutions, with a strong focus on performance, efficiency, and clean integration.

WORK EXPERIENCE

Embedded Software Engineer I, Henny Penny

Sept 2025 – Present

- Execute well-defined projects with a focus on development, testing, and troubleshooting.
- Contribute to system testing, validation, and documentation.
- Collaborating with cross-functional teams, supporting product development, and providing technical insights.
- Refactoring of system logic and maintaining a well-organized project environment.

Embedded System Developer, YenkoDev

Feb 2025 – Sept 2025

- Developing both hardware prototyping and software programming.

PROJECTS

- Addressing Illegal Parking in Damayan, Taytay Rizal A Vision-Based Alert System using YOLOv8 Algorithm and ByteTrack
- Automated PWD Parking Assistance System using YOLOv8 and EasyOCR
- Development of Power Consumption Meter with cloud data logging using PZEM, Arduino, and ESP32
- Hydroponics Project with Growth Monitoring using Image Processing
- Motorbanca Monitoring with Dashboard using LoRaWAN and ThingsBoard

EDUCATION

Bachelor of Engineering Technology Major in Electronics Technology (BET-ELXT)

2020 – 2024

Technological University of the Philippines – Taguig City

- Cum Laude

SKILLS

- **Programming Languages:** C, C++, Python, Bash, SQL, Java, Lua
- **MCUs:** ARM, AVR, Espressif, NRF52
- **Software Tools:** Git, Docker, CMake, Makefile, ESP-IDF
- **IDE:** STM32Cube, VSCode, MSVS, JetBrains
- **Hardware Tools:** Multimeter, Oscilloscope, Logic Analyzer
- **Protocols:** UART, SPI, I2C, I2S, CAN, RS232, RS485, LoRaWAN, MQTT, USB, BLE, WiFi
- **Debugger/Programmer:** JTAG, SWD, STLINK
- **Others:** FreeRTOS, PLC, IoT, Bootloaders